Jan 1, 2022

Day-30

TCP & UDP

# Functions of Layer 4 (Transport layer)

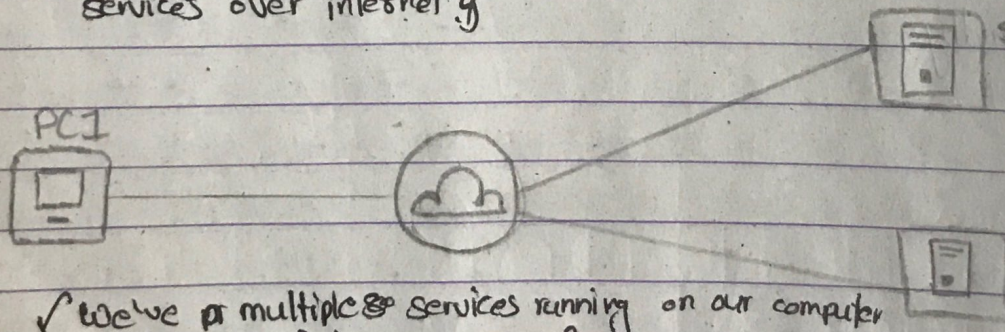
- Provides transparent transfer of data between end hosts.
- Provides (or doesn't provide) various services to applications:
  - reliable data transfer.
  - error recovery
  - data sequencing
  - flow control

- Provides Layer 4 addressing (port numbers). {Not physical interface/ports on network devices.}
  - ↳ Identify the Application layer protocol
  - ↳ Provides session multiplexing.

Session:

A session is an exchange of data between two or more communicating devices.

{ Imagine we've multiple internet tabs open, accessing different services over internet }




{ We've or multiple services running on our computer that need Internet Access }

In PC1 case, say its communicating with SRV1. At layer 4 its using TCP. It ~~uses src port of 50000~~

| PC1 | SRV1 |
|---|---|
| Src port: 50,000 | Src port: 80 |
| dst port: 80 | Dst: 50,000 |

When PC1 receives msg from SRV1, it the src. & dst. port numbers tell it that its the part of same communication session as the msg is sent earlier.

# Port numbers ranges

- ~~Provides transparent transfer of data between end hosts.~~
- The following ranges have been designated by IANA (Internet Assigned Numbers Authority).

1) # Well-known port numbers: 0-1023
Used for major protocols like HTTP, FTP etc & are very strictly regulated.

2) Registered port numbers: 1024-49151
Registration is required to use these port numbers although its not strict as well-known port numbers.

3) Ephemeral/private/dynamic port numbers: 49152-65535
They are also known as private or dynamic ports. Hosts (like PC) use these ports while selecting random source port.

NOB: Port numbers are fxn of both of main layer 4 protocols, TCP & UDP.

Scanned with CamScanner

# Transmission Control Protocol (TCP)

- TCP is connection oriented.
  - Before actually sending the data to destination host, the two hosts communicate to establish a connection. Once the connection is established, the data exchange begins.

- TCP provides reliable communication.
  - The destination host must knowledge that it received each TCP segment. (Segment = Layer 4 PDU like frame is of layer 2)
  - If a segment isn't acknowledged, it is sent again.

- TCP provides sequencing.
  - Sequence numbers in the TCP header allow destination hosts to put segments in the correct order even if they arrive out of order.

- TCP provides flow control.
  - The destination host can tell the source host to increase/decrease the rate data is sent.

## TCP Header

| ~~Offsets~~ Octet | ~~Octet~~ Octet | 0 | | | | | | | | 1 | | | | | | | | 2 | | | | | | | | 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Octet | Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 0 | 0 | Source Port | | | | | | | | | | | | | | | | Destination Port | | | | | | | | | | | | | | | |
| 4 | 32 | Sequence Number | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 8 | 64 | Acknowledge number (if ACK set) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 12 | 96 | Data offset | | | | Reserved | | | | N S | C R | E C | U R | A K | P S | R H | S T | Y N | F I N | Window Size | | | | | | | | | | | | | |
| 16 | 128 | Checksum | | | | | | | | | | | | | | | | | | Urgent Pointer (if URG set) | | | | | | | | | | | | | |
| 20 | 160 | Options (if data offset > 5. Padded at end with '0' bytes if necessary) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

| L2 Trailer | Data | L4 Header (TCP) | L3 Header | L2 Header |
|---|---|---|---|---|

- Source & Dst Port:-
  Each is 16 bits (2 bytes). So there are $2^{16}$ = 65,536 which is available port numbers. i.e <0-65,535>

- Sequencing & Ack Number.
  They provide sequencing & reliable communications.

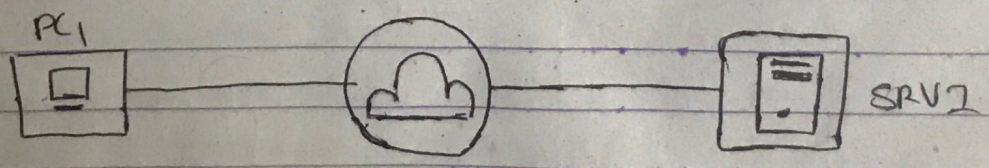
- Flags-
  TCP has series of 'Flags' bits which each serve a different purpose
  ACK, SYN & FIN are used to establish & terminate connection.

- Window size.
  It is used for flow control, adjusting the rate at which data is sent.

# Establishing connections: Three-Way Handshake.

TCP is connection oriented. The method TCP uses to establish connections is called TCP Three-Way Handshake.

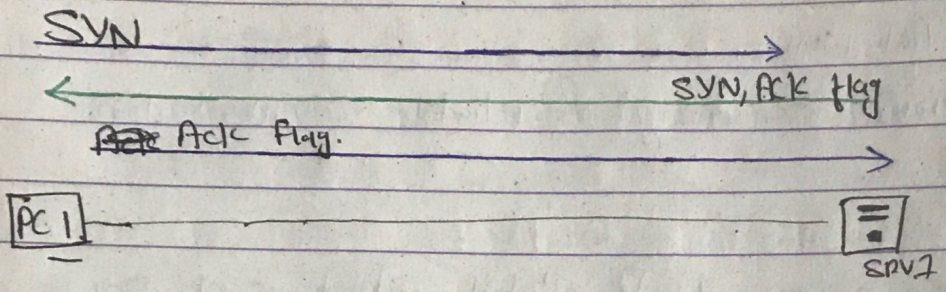
It is called three way because it involves three messages being sent between two hosts




So, lets say PC1 wants to access webpage on SRV1 using HTTP.

First it must establish a TCP connection (ACK & SYN)
To do so, it uses these two flags in the TCP header that could seen in table.

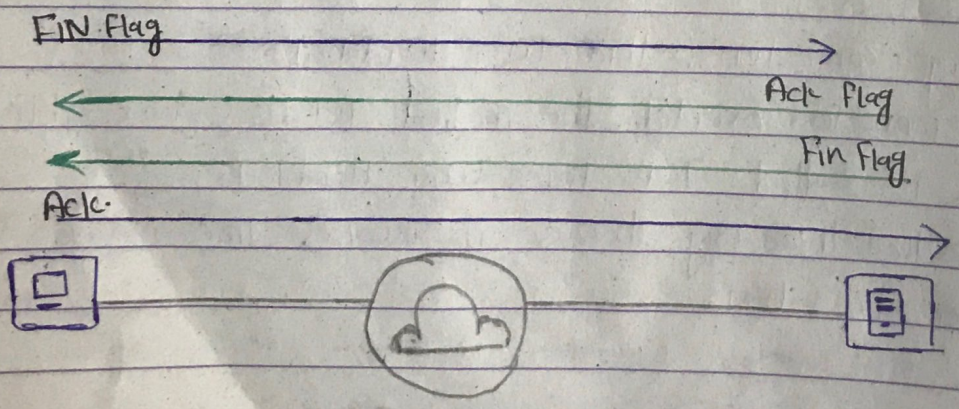
Ack → Acknowledgement
SYN → Synchronization




The real data exchange can then begin. The firs 3 msgs, are just to establish connection

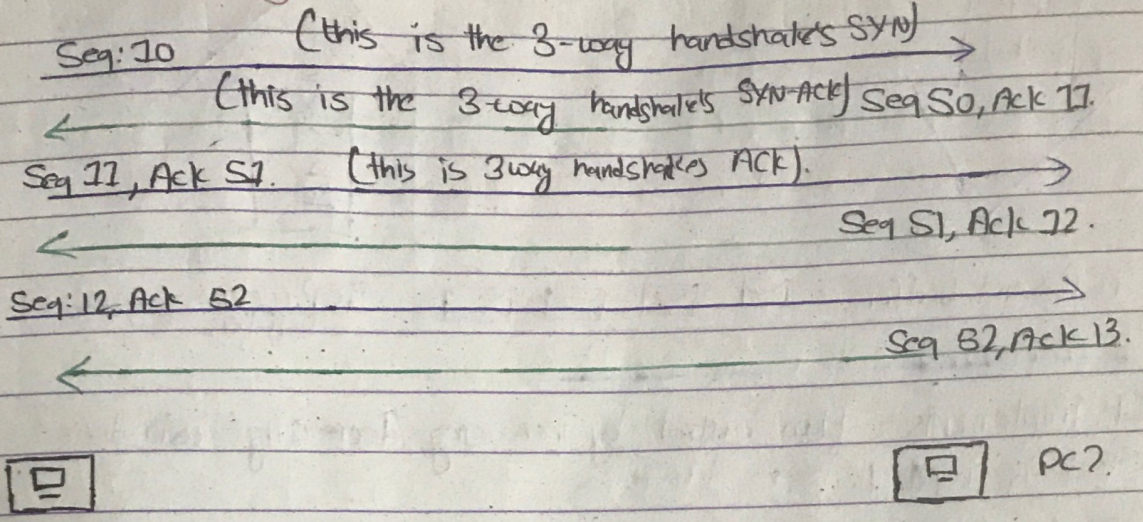
# Terminating Connections: Four-Way Handshake.

For terminating connection FIN & ACK flags are used.



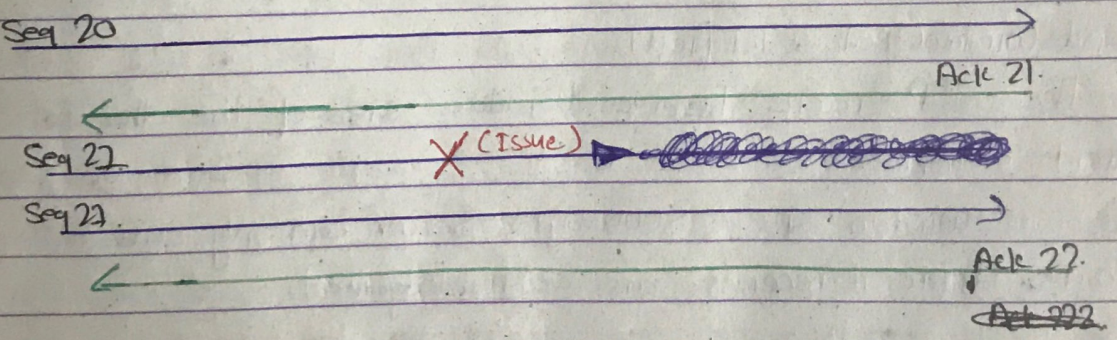

Scanned with CamScanner

# TCP: Sequencing / Acknowledgement




- Hosts set a random initial sequence number.
- Forward acknowledgement is used to indicate the seq. number of next segment the host expects to receive.

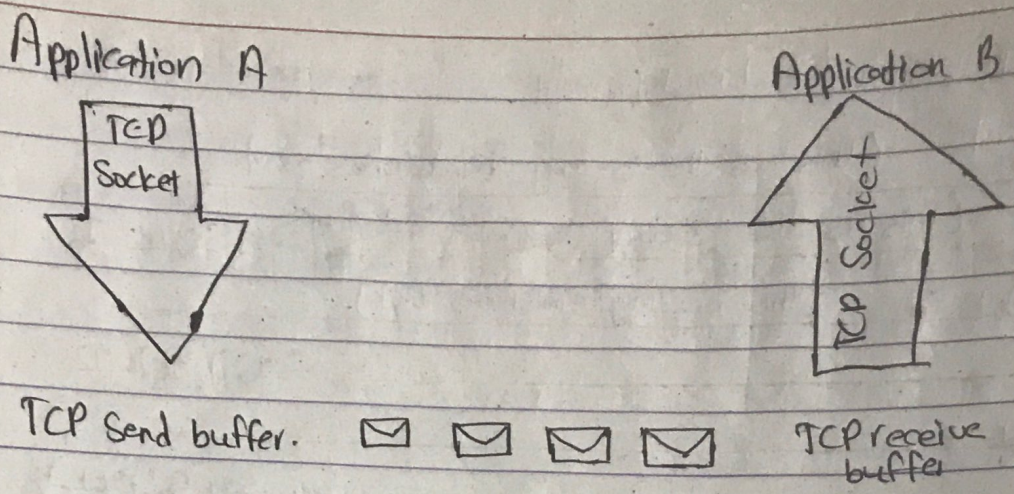
# TCP: Retransmission




# TCP Flow Control: Window Size

- Acknowledging every single segment, no matter what size, is inefficient.
- TCP flow control means that TCP will ensue that the sender is not overwhelming the receiver by sending more packets (segments) than it can handle.

Scanned with CamScanner




- TCP implements flow control by increasing / decreasing window size as required.
  Window size are variable during lifetime of connection

Receive Window.

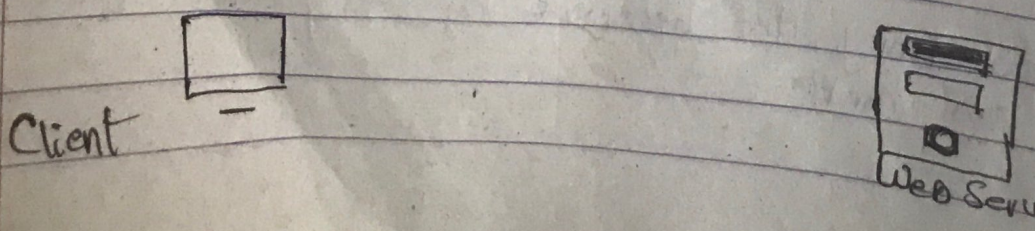
- Sending host can send only that amount of data before getting an acknowledgement & window update from this (the receiving) host.

Send window (not a TCP field)

The TCP receive Receive Windows size of the other host

Client Example

- Receive Window size - 5000 bytes - Server can only send 5000 bytes before it receives an acknowledgement.
- Send Window size - 10,000 bytes - Server told the client that it can send the server 10,000 bytes before receiving an acknowledgement.




Scanned with CamScanner

Web Server" - gp can send 5k bytes before receiving ACK and I can only receive 10k bytes at a time.

Server - Client's Send window - 5k. My Receive Window - 10k byte

Client - Server's Send window: 10k My Receive Window - 5k.

Send Window byte - This is the last byte that can be sent before receiving an ACK.

[For CCNA Just understand concept ].

# User Datagram Protocol (UDP)

- UDP is not connection-oriented (connectionless)

↳ The sending host doesn't establish a connection with the destination host before sending data. The data is simply sent.

- UDP doesn't provide reliable communication.

↳ When UDP is used, acknowledgements are not sent for received segments If a segment is lost, UDP has no mechanism to re-transmit it. Segments are sent best-effort.

- UDP doesn't provide sequencing.

There is no sequence number no. field in the UDP header. If segment arrive out of order, UDP has no mechanisms to put them back in order.

- UDP doesn't provide flow control

↳ UDP has no mechanism like TCP's window size to control flow of data.

Scanned with CamScanner

# UDP Header.

| Offsets | Octet | 0 | | | | | | | | 1 | | | | | | | | 2 | | | | | | | | 3 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Octet | Bit | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| 0 | 0 | Source Port | | | | | | | | | | | | | | | | Destination Port | | | | | | | | | | | | | | | |
| 4 | 32 | Length | | | | | | | | | | | | | | | | Checksum | | | | | | | | | | | | | | | |

# Comparing TCP & UDP.

- TCP provides more features than UDP, but at the cost of additional overhead.
- For applications that require reliable communications (for eg downloading a file), TCP is preferred.
- For applications like real-time voice and video, UDP is preferred.
- There are some applications that use UDP, provide reliability etc. within application itself.
- Some applications use both TCP & UDP, depending on situation.

| TCP | UDP |
|---|---|
| Connection-Oriented | Connectionless |
| Reliable | Unreliable |
| Sequencing | No-sequencing |
| Flow control | No-flow control |
| Use for downloads, file sharing etc. | Used for VoIP, live video, etc. |

Scanned with CamScanner

# Port Numbers

| TCP | UDP | TCP & UDP |
| --- | --- | --- |
| FTP data (20) | DHCP server (67) | DNS (53) |
| FTP control (21) | DHCP Client (68) | |
| SSH (22) | TFTP (69) | |
| Telnet (23) | SNMP agent (161) | |
| SMTP (25) | SNMP manager (162) | |
| HTTP (80) | Syslog (514) | |
| POP3 (110) | | |
| HTTPS (443) | | |
| IMAP (143) | | |

Scanned with CamScanner